

Probability and Random Processes

Monsoon 2026

Exam: Quiz 1
Total Marks: 25

Date: 08 Sept, 2026
Time: 11:45 am to 01:00 pm

Instructions:

- All questions are compulsory.
- Clearly state the assumptions (if any) made that are not specified in the question.
- Calculators are allowed.

1. Answer the following questions in short. [3x5]

1. Explain with an example why mutual exclusivity of two events does not imply their independence.
2. Define cumulative distribution function (CDF) and state its properties.
3. Consider $X \sim \text{Binomial}(N, p)$. Find the conditional probability mass function of X given that $X > n$.
4. Consider a fair coin is tossed for three times. Find out the conditional probability of the number of heads is more than the number of tails given that first toss outcome is head.
5. Consider X is a continuous random variable whose pdf is given by

$$f_X(x) = \begin{cases} k \cdot x(6-x) & \text{for } 0 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

Find out k , obtain the mean of X , and provide your observation.

2. Show that

$$\mathbf{E}[X] = \mathbf{E}_Y[\mathbf{E}_X[X|Y]]$$

[5]

3. Suppose that X and Y are independent, identically distributed, geometric random variables with parameter p . Show that [5]

$$\mathbf{P}[X = i | X + Y = n] = \frac{1}{n-1}$$

and give your observations.

ALL THE BEST!

Q: Explain with an example why mutual exclusivity of 2 events does not imply their independence.

Sol: ① Define Mutual exclusivity of 2 events with valid example.
2 events A & B are said to be mutually exclusive if $A \cap B = \emptyset$
or $P(A \cap B) = 0$ or when both events cannot occur simultaneously. (0.5M)

② Eg: Getting Head and Tail in one Toss. (I)
 $P(H) = \frac{1}{2}$, $P(T) = \frac{1}{2}$ and $P(H \cap T) = 0 \Rightarrow$ Mutual exclusive events. (1M)

③ Show why they are not independent:

Use $P(A \cap B) = P(A)P(B)$ for independence and show that equality fails (0.5M)

Eg: From (I) $P(H) = \frac{1}{2}$, $P(T) = \frac{1}{2} \Rightarrow P(H) \cdot P(T) = \frac{1}{4} \neq P(H \cap T)$
 $\Rightarrow H \& T$ are independent. — (I) (1M)

From (I) and (II) we can say that mutually exclusivity of 2 events does not imply their independence.

Q1.2**[3 marks]****Definition of CDF:****[0.75 mark]**

A cumulative distribution function (CDF) of a random variable X gives the probability that X takes a value less than or equal to x . It is defined as

$$F_X(x) = P(X \leq x), \quad -\infty < x < \infty.$$

Marking:

- Correct explanation/definition in words: 0.50 mark.
- Correct mathematical expression: 0.25 mark.
- If only $F_X(x) = P(X \leq x)$ is written without an explanation, give 0.50 marks.

Properties of CDF:**[Any $3 \times 0.75 = 2.25$ marks]**

Students may state **any three** of the following properties.

1. Boundedness:

$$0 \leq F_X(x) \leq 1, \quad \forall x \in \mathbb{R}.$$

2. Non-decreasing :

$$x_1 < x_2 \implies F_X(x_1) \leq F_X(x_2).$$

3. Right-continuity:

$$\lim_{h \rightarrow 0^+} F_X(x+h) = F_X(x).$$

4. Limit at $-\infty$:

$$\lim_{x \rightarrow -\infty} F_X(x) = 0.$$

5. Limit at $+\infty$:

$$\lim_{x \rightarrow +\infty} F_X(x) = 1.$$

6. Interval probability:

$$P(a < X \leq b) = F_X(b) - F_X(a), \quad a < b.$$

7. Point probability / jump property:

$$P(X = k) = F_X(k) - F_X(k^-).$$

Marking notes:

- Award 0.75 mark for each correct property.
- If a student writes more than three properties, marks given for **only the first three properties written**. Later properties must not be used to replace an incorrect property among the first three.
- “Monotonically increasing” is not accepted for “non-decreasing”.

Total = 0.75 + 3(0.75) = 3 marks

Solutions and Grading Rubrics

Question 1.3

Problem: Consider $X \sim \text{Binomial}(N, p)$. Find the conditional probability mass function of X given that $X > n$.

Solution

To find the conditional probability mass function (PMF) $P(X = k \mid X > n)$, we apply the definition of conditional probability:

$$P(X = k \mid X > n) = \frac{P(X = k \text{ and } X > n)}{P(X > n)}$$

1. Support of the Distribution: Since we are given that $X > n$, the possible values that k can take are $k = n + 1, n + 2, \dots, N$. For any $k \leq n$, the probability is 0.

2. Denominator Calculation $P(X > n)$: The probability that X exceeds n is given by summing the binomial PMF over the valid range:

$$P(X > n) = \sum_{j=n+1}^N \binom{N}{j} p^j (1-p)^{N-j}$$

3. Numerator Calculation $P(X = k \text{ and } X > n)$: For any $k \in \{n + 1, n + 2, \dots, N\}$, the event $X = k$ inherently satisfies $X > n$. Therefore:

$$P(X = k \text{ and } X > n) = P(X = k) = \binom{N}{k} p^k (1-p)^{N-k}$$

4. Final Conditional PMF: Combining the numerator and denominator, the conditional PMF is:

$$P(X = k \mid X > n) = \begin{cases} \frac{\binom{N}{k} p^k (1-p)^{N-k}}{\sum_{j=n+1}^N \binom{N}{j} p^j (1-p)^{N-j}}, & \text{for } k = n + 1, n + 2, \dots, N \\ 0, & \text{otherwise} \end{cases}$$

Grading Rubric

- **1 Mark:** Stating the conditional probability definition $P(X = k \mid X > n) = \frac{P(X=k)}{P(X>n)}$ OR correctly specifying the support range $k = n + 1, \dots, N$.
- **1 Mark:** Correct expression for the binomial term in the numerator $\binom{N}{k} p^k (1-p)^{N-k}$ AND the sum/complement setup for the denominator $P(X > n)$.
- **1 Mark:** Combining into the correct final ratio expression for the conditional PMF.

Question 1.4

Problem: Consider a fair coin is tossed three times. Find out the conditional probability of the number of heads is more than the number of tails given that first toss outcome is head.

Solution

1. Sample Space and Event Definitions: When a fair coin is tossed 3 times, the sample space S contains $2^3 = 8$ equally likely outcomes:

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

Define the following events:

- **Event A:** Number of heads > Number of tails (i.e., at least 2 heads out of 3 tosses).

$$A = \{HHH, HHT, HTH, THH\}$$

- **Event B:** First toss outcome is a Head.

$$B = \{HHH, HHT, HTH, HTT\} \implies |B| = 4$$

2. Favorable Intersection Outcomes ($A \cap B$): The outcomes where the first toss is a head AND heads outnumber tails:

$$A \cap B = \{HHH, HHT, HTH\} \implies |A \cap B| = 3$$

3. Conditional Probability Calculation: Using the reduced sample space or conditional probability formula:

$$P(A | B) = \frac{|A \cap B|}{|B|} = \frac{3}{4} = 0.75$$

Grading Rubric

- **1 Mark:** Identifying the conditioning event/reduced sample space (stating $|B| = 4$ or listing the 4 outcomes starting with Head).
- **1 Mark:** Identifying the favorable outcomes where heads outnumber tails given first toss is Head ($\{HHH, HHT, HTH\}$, count = 3).
- **1 Mark:** Correct calculation yielding the final answer of $\frac{3}{4}$ (or 0.75).

1.5

Given pdf, $f_x(u) = \begin{cases} k \cdot x(6-x), & \text{for } 0 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$, Find $k, E(x)$, observations.

from properties of pdf, we know that,

$$\int_0^x f_x(u) dx = 1$$

$$\int_0^x k \cdot (6x - x^2) dx = 1$$

$$k \left(6 \left(\frac{x^2}{2} \right) - \left(\frac{x^3}{3} \right) \right) = 1$$

$$k \left(6 \cdot \frac{6^2}{2} - \frac{6^3}{3} \right) = 1$$

$$k \left(6^3 \left(\frac{1}{2} - \frac{1}{3} \right) \right) = 1$$

$$k \left(6^3 \cdot \frac{1}{6} \right) = 1$$

$$k = \frac{1}{36}$$

→ IM

$$\text{Mean} = E(x) = \int_0^x f_x(u) \cdot u du$$

$$E(x) = \int_0^6 k (6x - x^2) \cdot x dx$$

$$= \int_0^6 k (6x^2 - x^3) dx$$

$$= k \left(6 \frac{x^3}{3} - \frac{x^4}{4} \right) \Big|_0^6$$

$$= k \left(6 \cdot \frac{6^3}{3} - \frac{6^4}{4} \right)$$

$$= k \left(6^4 \left(\frac{1}{3} - \frac{1}{4} \right) \right)$$

$$= \frac{1}{36} \left(6^4 \left(\frac{1}{12} \right) \right) = \frac{6^2}{12} = 3$$

$$E(x) = 3$$

→ IM

substituting k in pdf at each x ,

$$\text{at } x=0, f_x(u) = k \cdot 0 \cdot (6) = 0$$

$$\text{at } x=1, f_x(u) = k \cdot 1(5) = \frac{1}{36} (5) = \frac{5}{36} = 0.13$$

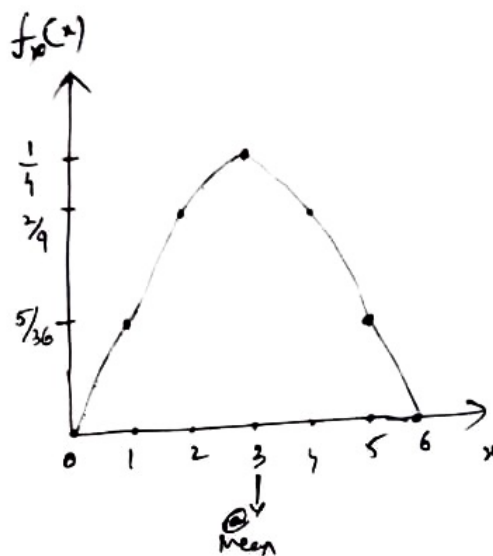
$$\text{at } x=2, f_x(u) = k \cdot 2(4) = \frac{1}{36} (8) = \frac{2}{9} = 0.22$$

$$\text{at } x=3, f_x(u) = k \cdot 3(3) = \frac{1}{36} (9) = \frac{1}{4} = 0.25$$

$$\text{at } x=4, f_x(u) = k \cdot 4(2) = \frac{1}{36} (8) = \frac{2}{9} = 0.22$$

$$\text{at } x=5, f_x(u) = k \cdot 5(1) = \frac{1}{36} (5) = \frac{5}{36} = 0.13$$

$$\text{at } x=6, f_x(u) = k \cdot 6(0) = 0$$



Observations :-

① Pdf is symmetric about the mean i.e at $x=3$

② The mean is located at halfway down the interval. ($0 \leq x \leq 6$) ($x=3$)

③ Any other ^{Valid} Observation which you have written

IM
(1 out of 3 → 0.5M)
≥ 2 out of 3 → 1M

Question 2: Law of Total Expectation

[5 Marks]

Show that

$$E[X] = E_Y [E[X | Y]]$$

A. Discrete Random Variables

Suppose X and Y are discrete random variables.

Step 1: Write the conditional expectation

[1 Mark]

For a particular value $Y = y$,

$$E[X | Y = y] = \sum_x x P(X = x | Y = y).$$

Step 2: Take expectation over Y

[1 Mark]

$$\begin{aligned} E_Y [E[X | Y]] &= \sum_y E[X | Y = y] P(Y = y) \\ &= \sum_y \left[\sum_x x P(X = x | Y = y) \right] P(Y = y). \end{aligned}$$

Step 3: Convert conditional probability to joint probability

[1 Mark]

Using

$$P(X = x | Y = y) P(Y = y) = P(X = x, Y = y),$$

we obtain

$$E_Y [E[X | Y]] = \sum_y \sum_x x P(X = x, Y = y).$$

Step 4: Marginalization / Law of Total Probability

[1 Mark]

Rearranging the summation,

$$E_Y [E[X | Y]] = \sum_x x \left[\sum_y P(X = x, Y = y) \right].$$

Using marginalization,

$$\sum_y P(X = x, Y = y) = P(X = x),$$

we get

$$E_Y [E[X | Y]] = \sum_x x P(X = x).$$

Important grading note: The marginalization step (or an equivalent use of the law of total probability) must be explicitly shown or mentioned. If a student directly jumps to the final result without this step, deduct 1 mark.

Step 5: Recognize the expectation of X [1 Mark]

By definition,

$$E[X] = \sum_x xP(X = x).$$

Therefore,

$$E_Y [E[X | Y]] = E[X].$$

Hence,

$$E[X] = E_Y [E[X | Y]].$$

B. Continuous Random Variables

Suppose X and Y are continuous random variables with joint probability density function $f_{X,Y}(x, y)$.

Step 1: Write the conditional expectation [1 Mark]

For a particular value $Y = y$,

$$E[X | Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx.$$

Step 2: Take expectation over Y [1 Mark]

$$\begin{aligned} E_Y [E[X | Y]] &= \int_{-\infty}^{\infty} E[X | Y = y] f_Y(y) dy \\ &= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x f_{X|Y}(x | y) dx \right] f_Y(y) dy. \end{aligned}$$

Step 3: Convert conditional density to joint density [1 Mark]

Using

$$f_{X|Y}(x | y) f_Y(y) = f_{X,Y}(x, y),$$

we obtain

$$E_Y [E[X | Y]] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{X,Y}(x, y) dx dy.$$

Step 4: Marginalization**[1 Mark]**

Rearranging the integrals,

$$E_Y [E[X | Y]] = \int_{-\infty}^{\infty} x \left[\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \right] dx.$$

Using marginalization,

$$\int_{-\infty}^{\infty} f_{X,Y}(x, y) dy = f_X(x),$$

we get

$$E_Y [E[X | Y]] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Important grading note: The marginalization step must be explicitly shown or mentioned. If a student directly jumps to the final result without this step, deduct 1 mark.

Step 5: Recognize the expectation of X **[1 Mark]**

By definition,

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

Therefore,

$$\boxed{E_Y [E[X | Y]] = E[X]}.$$

Hence,

$$\boxed{E[X] = E_Y [E[X | Y]]}.$$

Marking Scheme — 5 Marks

The following marking scheme applies to either the discrete or continuous derivation.

Step	Marks
1. Correct definition of $E[X Y = y]$	1
2. Correctly taking expectation with respect to Y	1
3. Correctly using the conditional-to-joint relationship	1
4. Explicitly performing marginalization over Y (or equivalent use of the law of total probability)	1
5. Recognizing the definition of $E[X]$ and stating the conclusion	1
Total	5

Note

A student is not required to prove both the discrete and continuous cases unless the question explicitly asks for both.

A complete and correct derivation for either the discrete or continuous case should receive full marks.

Ques 3

Given X, Y are iid RV (Independent, Identical R.V)

$$\Rightarrow P_{xy}[x, y] = P_x(x) \cdot P_y(y)$$

Given that X, Y follows Geometric Distribution with parameter p

$$P_x(x) = P[X=x] = p q^{x-1}; x=1, 2, \dots, \infty \quad \text{where } [q=1-p]$$

$$P_y(y) = P[Y=y] = p q^{y-1}; y=1, 2, \dots, \infty \rightarrow 0.5 \text{ Marks}$$

$$\text{Now, } P_{xy}[x, y] = P[X=x, Y=y] = P_x(x) \cdot P_y(y) \quad \left\{ \begin{array}{l} \text{Independent} \end{array} \right\}$$

$$\Rightarrow P_{xy}[x, y] = p q^{x-1} \cdot p q^{y-1} = p^2 q^{x+y-2}; \begin{array}{l} x=1, \dots, \infty \\ y=1, \dots, \infty \end{array} \rightarrow \textcircled{1}$$

$$\text{Now, } P[X=i | X+Y=n] = \frac{P[X=i \wedge X+Y=n]}{P[X+Y=n]} \quad \left\{ \begin{array}{l} \text{Conditional Law} \\ P(A|B) = \frac{P(A \wedge B)}{P(B)} \end{array} \right. \rightarrow 0.5 \text{ Mark}$$

Now we have $\rightarrow$

$$P[X=i \wedge X+Y=n] = P[X=i, Y=n-i] \Rightarrow p^2 q^{(i+n-i-2)} = p^2 q^{n-2} \rightarrow \textcircled{2}$$

as $X=i$ and $X+Y=n \uparrow$

1 Mark { from eq ① }

Also $P[X+Y=n] \Rightarrow$ Since X starts from 1 and goes till ∞ but Y also starts from 1 and goes till ∞ i.e. $X \geq 1$
 $Y \geq 1$

$$X=1 \quad Y=n-1$$

$$X=2 \quad Y=n-2$$

⋮

$$X=n-1 \quad Y=1$$

Other values are Not Possible of (x, y)
 Such that $X+Y=n$.

Therefore, $P[X+Y=n] = \sum_{i=1}^{n-1} P[X=i, Y=n-i] = \sum_{i=1}^{n-1} p^2 q^{(i+n-i-2)}$

$$P[X+Y=n] = \sum_{i=1}^{n-1} p^2 q^{n-2} = p^2 q^{n-2} \sum_{i=1}^{n-1} 1 = p^2 q^{n-2} (n-1)$$

1 Mark

→ (3)

now from Eq (2) and (3)

$$P[X=i | X+Y=n] = \frac{p^2 q^{n-2}}{p^2 q^{n-2} (n-1)} = \frac{1}{(n-1)} = \text{Ans}$$

1 Mark

Observation: → This Conditional Probability does not depend on the Success parameter p and the specific trial i , implies that the distribution is discrete uniform distribution over the set $\{1, 2, \dots, n-1\}$

1 Mark

Also, ~~Another way~~ to think; $P[X=i | X+Y=n]$

we need n -trials to reach two success ($X+Y=n$)

therefore the second last trial should be second success therefore the first trial will be equally likely to have occurred on any of the $(n-1)$ trials.